Checkpoints Answers

7.6 a) ions present: Cu^{2+} , SO_4^{2-} , Na^+ , OH^-

equation: $Cu^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Cu(OH)_{2(s)}$

b) ions present: Na^+ , S^{2-} , Ca^{2+} , NO_3^-

equation: $Ca^{2+}_{(aq)} + S^{2-}_{(aq)} \rightarrow CaS_{(s)}$

8.1 a) i) $35.5/12 \approx 3$ iii) $47.9/12 \approx 4$
ii) $107.9/12 \approx 9$ iv) $83.8/12 \approx 7$

b) i) $4.0/1.0 = 4$ iii) $12.0/1.0 = 12$
ii) $16.0/1.0 = 16$ iv) $55.8/1.0 = 55.8$

c) $107.9/27.0 \approx 4$

d) i) Na, Mg, Al ii) H, He, B

e) Ca

8.2 a) $M_r (CO_2) = (1 \times 12.0) + (2 \times 16.0) = 44$

b) $M_r (Cl_2) = (2 \times 35.5) = 71$

c) $M_r (CuSO_4) = 63.5 + 32.1 + (4 \times 16.0)$
$= 159.6$

d) $M_r (Pb(NO_3)_2)$
$= 207.2 + (2 \times 14.0) + (6 \times 16.0) = 331.2$

e) $M_r (Mg(OH)_2)$
$= 24.3 + (2 \times 16.0) + (2 \times 1.0) = 58.3$

8.3 a) $M_r (SO_2) = 32.1 + (2 \times 16.0) = 64.1$

b) $M_r (AlCl_3) = 27.0 + (3 \times 35.5) = 133.5$

c) $M_r (Cl_2) = (2 \times 35.5) = 71.0$

d) $M_r (NH_3) = 14.0 + (3 \times 1.0) = 17.0$

e) $M_r (BaCl_2) = 137.3 + (2 \times 35.5) = 208.3$

f) $M_r ((NH_4)_2SO_4) = (2 \times 14.0) + (8 \times 1.0)$
$+ 32.1 + (4 \times 16.0) = 132.1$

g) $M_r (Fe(OH)_2)$
$= 55.8 + (2 \times 16.0) + (2 \times 1.0) = 89.8$

8.4 a) %H (by mass) $= (6.0/46) \times 100 = 13.0\%$

b) %O (by mass) $= (16.0/46) \times 100 = 34.8\%$

8.5 a) $M_r (CO_2) = 12.0 + (2 \times 16.0) = 44.0$
% C $= (12.0 / 44) \times 100 = 27.3\%$
% O $= (32.0 / 44) \times 100 = 72.7\%$

b) $M_r (PbSO_4) = 207.2 + 32.1 + (4 \times 16.0)$
$= 303.3$
% Pb $= (207.2 / 303.3) \times 100 = 68.3\%$
% S $= (32.1 / 303.3) \times 100 = 10.6\%$
% O $= (64.0 / 303.3) \times 100 = 21.1\%$

8.6 a) $M_r (AgNO_3) = 107.9 + 14.0 + (3 \times 16.0)$
$= 169.9$
% Ag $= (107.9 / 169.9) \times 100 = 63.5\%$
% N $= (14.0 / 169.9) \times 100 = 8.2\%$
% O $= (48.0 / 169.9) \times 100 = 28.3\%$

b) $M_r (Na_2SO_4)$
$= (2 \times 23.0) + 32.1 + (4 \times 16.0) = 142.1$
% Na $= (46.0 / 142.1) \times 100 = 32.4\%$
% S $= (32.1 / 142.1) \times 100 = 22.6\%$
% O $= (64.0 / 142.1) \times 100 = 45.0\%$

c) $M_r ((NH_4)_2SO_4)$
$= (2 \times 14.0) + (8 \times 1.0) + 32.1 + (4 \times 16.0)$
$= 132.1$
% N $= (28 / 132.1) \times 100 = 21.2\%$
% H $= (8.0 / 132.1) \times 100 = 6.1\%$
% S $= (32.1 / 132.1) \times 100 = 24.3\%$
% O $= (64.0 / 132.1) \times 100 = 48.4\%$

9.1 The mass of copper atoms is greater than the mass of carbon atoms. The mass of Cu atoms is $63.5/12.0$ (≈ 5.3) times greater.

9.2 A mole of 5¢ coins $= 6.02 \times 10^{23} \times 0.05$ dollars $\approx \$3 \times 10^{22}$ dollars. This is much, much more than a million (10^6).

9.3 a) 6.02×10^{23} O atoms

b) 1.204×10^{24} O atoms

c) 1.204×10^{24} O atoms

d) 6.02×10^{24} O atoms

e) 1.806×10^{25} O atoms

9.4 a) 64.1 g mol^{-1} – see text

b) $23.0 + 35.5 = 58.5$ g mol^{-1}

c) $(2 \times 1.0) + 16.0 = 18.0$ g mol^{-1}

d) $65.4 + (2 \times 35.5) = 136.4$ g mol^{-1}

e) $(2 \times 27.0) + (3 \times 16.0) = 102$ g mol^{-1}

Set 12: Relative atomic mass

1. a) nitrogen, oxygen, neon

b) carbon, aluminium, chlorine

c) chromium, iron, zinc

2. a) titanium

b) krypton

c) iodine

3. a) 16.0/4.0 = 4

b) 107.9/10.8 ≈ 10

c) 12.0/1.0 = 12

4. argon (40) and calcium (40.1)

5. The presence of different isotopes for a particular element means that the average relative atomic mass for that element may not be a whole number e.g. chlorine has two isotopes, chlorine-35 (≈ 75%) and chlorine-37 (≈ 25%) giving an average A_r of 35.5.

6. Since we are dealing with *relative* masses the new definition suggested by the chemists would work equally as well. Also since $^1/_{16}$ of the mass of an oxygen-16 atom is the same as $^1/_{12}$ the mass of a carbon-12 atom, the relative masses of the elements as listed on Table 1 would not be affected.

Set 13: Relative molecular mass

1. a) M_r (N_2) = 28

b) M_r (Cl_2) = 71

c) M_r (CO_2) = 44

d) M_r (SO_2) = 64.1

e) M_r (SO_3) = 80.1

f) M_r (CH_4) = 16

2. a) M_r (NaCl) = 23.0 + 35.5 = 58.5

b) M_r (CaO) = 40.1 + 16 = 56.1

c) M_r ($AlCl_3$) = 27.0 + (3 x 35.5) = 133.5

d) M_r (ZnS) = 65.4 + 32.1 = 97.5

e) M_r (MgO) = 24.3 + 16.0 = 40.3

f) M_r (KOH) = 39.1 + 16.0 + 1.0 = 56.1

3. a) M_r ($Zn(NO_3)_2$)
= 65.4 + (2 x 14.0) + (6 x 16.0) = 189.4

b) M_r ($Al_2(SO_4)_3$)
= (2 x 27.0) + (3 x 32.1) + (12 x 16) = 342.3

c) M_r ($(NH_4)_2CO_3$)
= (2x14.0) + (8 x 1.0) + 12.0 + (3 x 16.0) = 96.0

d) M_r ($Mg(OH)_2$)
= 24.3 + (2 x 16.0) + (2 x 1.0) = 58.3

e) M_r ($CuCO_3$)
= 63.5 + 12.0 + (3 x 16.0) = 123.5

f) M_r (FeO) = 55.8 + 16.0 = 71.8

4. a) O_2 b) SO_2 c) I_2 d) SO_3

5. M_r (H_2O) = 18 , M_r (CO) = 28 ,
M_r (CH_3COOH) = 60, M_r (C_2H_5OH) = 46 ,
M_r (NH_3) = 17 , M_r (C_3H_8) = 44

a) 5 C atoms = 60 \ same as CH_3COOH

b) 17 H atoms = 17 \ same as NH_3

6. Not all substances are made up of molecules and so it would be incorrect to use the term relative *molecular* mass in all cases.
For substances such as ionic compounds (which are made up of an infinite array of ions) we refer to their relative *formula* mass.

Set 14: Percentage composition

1. a) M_r (SO_2) = 32.1 + (2 x 16.0) = 64.1
% S = (32.1 / 64.1) x 100 = 50.1%
% O = (32.0 / 64.1) x 100 = 49.9%

b) M_r (CO) = 12.0 + 16.0 = 28
% C = (12 / 28) x 100 = 42.9%
% O = (16 / 28) x 100 = 57.1%

c) M_r (NaCl) = 23.0 + 35.5 = 58.5
% Na = (23 / 58.5) x 100 = 39.3%
% Cl = (35.5 / 58.5) x 100 = 60.7%

d) M_r ($Mg(OH)_2$)
= 24.3 + (2 x 16.0) + (2 x 1.0) = 58.3
% Mg = (24.3 / 58.3) x 100 = 41.7%
% O = (32.0 / 58.3) x 100 = 54.9%
% H = (2.0 / 58.3) x 100 = 3.4%